

TPC-MVP4: A Source-Locked Route Decision after TPC-133–141: Frontier Totalization, Arithmetic Shadows, and the Strict Endpoint

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Abstract

TPC-MVP3 reported the requirement for a complete native archive as the first missing object in a proposed fixed-shift prime-pair route. TPC-133–136 now make the native entrance executable and follow it through an exact declared dyadic cut. They reconnect the eligible-prefix, eligible-tail and frontier terminal classes to the original packet, prove the eligible prefix soft, and give a formal support-envelope path-matrix witness that the eligible-only TPC-17/18 compiler is coefficientwise incomplete. The witness coefficient itself may vanish. This cut completeness is not full-carrier totalization. TPC-137–140 separately prove a fixed-data logarithmic arithmetic shadow, a small-polylog affine power-of-log estimate outside a small logarithmic-density exceptional set, and two scoped non-transfer firewalls, then formulate the remaining actual-family and selector estimates needed by the physical determinant-two packet.

We read these results only through the source-locked TPC-141 manifest. Hash equality is only a drift/integrity check; it does not prove a source theorem or generator. Invalid hashes, scopes, carriers, normalizations, terminal covers, occurrence registries or endpoint records return no mathematical verdict. A valid snapshot is classified by seven ordered outcomes:

GO, ARCHITECTURE-INFEASIBLE, REROUTE,
STOP-ROUTE, NOT-TESTABLE, ARITHMETIC-FRONTIER, OPEN.

The arithmetic-frontier label requires all selected-route structural nodes and the strict physical 1/400 endpoint to be proved independently of the unresolved arithmetic targets. Fixed, logarithmic, almost-scale, conditional and scope-mismatched results cannot trigger it.

The small-polylog result is a restricted external positive arithmetic shadow, not positive L2: actual CRT-family containment, squarefree/periodic reassembly, local exceptional-set control on every terminal window and deterministic prefixes are not proved. The frozen snapshot is valid, but its first missing node is now **H1.frontier_totalization**. The frontier has neither complete downstream physical maps nor a complete original-scale $o(X)$ theorem; H6–H9 remain untestable; and no positive L2 estimate occurs. Hence

Verdict_{133:141} = NOT-TESTABLE.

This is genuine movement of the first-failure boundary, not GO, not the arithmetic frontier, and not a negative theorem about the twin-prime conjecture.

Keywords: fixed shift; source-locked snapshot; route decision; unsupported carrier; arithmetic frontier; parity barrier.

Snapshot boundary. This decision is dated 27 July 2026 and applies only to the declared TPC-MVP1 architecture, the selected frontier-augmented signed route, and the exact TPC-141 source manifest. A changed source hash, route, scope, carrier, normalization or occurrence registry creates a new snapshot; an unchanged hash supplies integrity, not mathematical validity. No conclusion here asserts a prime-pair theorem or an impossibility result for twin primes.

1 What changed after MVP3

MVP1 resolved a conditional proof program into H1–H9 [3]. MVP2 separated proved interfaces from missing growing inputs [4]. MVP3 added a strict arithmetic-frontier label and reported H1.archive as the first missing record [6]. TPC-131 supplied the general composition formula

$$|B_{h_0, \delta}(X)| \leq o(X) + X^{1 - \sigma_{\text{raw}} + \Lambda_{\text{phys}} + o(1)}$$

under a complete proof-carrying archive and occurrence registry [5].

TPC-133–136 provide more than another abstract schema. They enumerate the native entrance in an effective support envelope, create a coefficientwise conservative dyadic path archive, partition the TPC-17/18 eligible domain from its frontier, and reconnect the three declared cut-terminal classes exactly [7–10]. Thus the old generic archive absence has been replaced by a precise downstream totality defect.

The arithmetic branch also becomes sharper but not stronger in the required sense. TPC-137 proves a frozen fixed-affine logarithmic closure based on the formal two-point theorem of Tao [2, 11]. TPC-138 records the quantitative shift-one scope of Helfgott–Radziwiłł and proves that the direct reparameterization is blocked on the active odd determinant-two carrier [1, 12]. TPC-138/139 also record the Tao–Teräväinen small-polylog affine almost-scale theorem as a restricted positive shadow. TPC-139 and TPC-140 then expose, rather than assume, the remaining actual-family containment, squarefree/periodic reassembly, local exceptional-window and actual-prefix selector gates [13, 14]. In particular, a cumulative logarithmic-density estimate is not promoted to the same normalized estimate on every terminal window.

2 A frozen source-locked snapshot

Definition 2.1 (Snapshot validity). An MVP4 snapshot is valid only if:

- (i) the TPC-141 manifest hash and every TPC-133–140 source bundle hash agree with the frozen files;
- (ii) node, export, import, scope, carrier and normalization tables are internally consistent;
- (iii) the proof DAG is acyclic and its declared topological order exactly covers the nodes;
- (iv) the three cut terminal roles and four residual namespaces are exact covers;
- (v) the occurrence registry has unique identifiers, acyclic nonoverlapping joint replacements and no missing primitive leaf;
- (vi) the selected route and every stop record belong to the declared route universe; and
- (vii) the endpoint uses exact rational direction: upper certificates pass, lower certificates stop, and unknown values remain unknown.

An invalid snapshot returns INVALID-SNAPSHOT, which is a data-validation outcome rather than a route verdict.

Remark 2.2 (Hash scope). A content hash detects source drift. It does not prove a theorem, the completeness of a generator, or the uniqueness of a tuple key. Those claims require the source proofs and raw tuple validation.

3 Statuses, evidence and active requirements

Node status and program evidence remain orthogonal:

PROVED, CONDITIONAL, OPEN, NOT-TESTABLE, REFUTED.

An L0 record is a finite or abstract identity. An L1 record is attached to the literal fixed- h_0 carrier and normalization. A positive L2 record must be a uniform growing power estimate on the complete actual coefficient family with all masks, weights, phases, origins, prefixes and residual sectors.

Accordingly:

- fixed-data or fixed-periodic logarithmic cancellation is an arithmetic shadow, not positive L2;
- a small-polylog affine power-of-log estimate outside an exceptional scale set is a restricted positive arithmetic shadow, not actual-family containment, a local terminal-window estimate, or deterministic-prefix control;
- a shift-one estimate is not an estimate on a scope-mismatched determinant-two progression;
- a non-effective slow diagonal is not the prescribed physical schedule;
- almost-all-scale cancellation is not all-prefix control; and
- a conditional power-ledger formula is not a certified positive exponent.

Route requirements are an explicit selected AND/OR graph. A stopped positive or eligible-only sufficient subroute need not be proved on a selected signed alternative. Conversely, one arithmetic theorem cannot cover two packet classes unless a joint exact-cover theorem says so.

4 Eight total audit outcomes

The outer audit has one validation outcome and seven mathematical route verdicts.

Definition 4.1 (Ordered valid-snapshot verdicts). After [definition 2.1](#) passes, test:

- (i) GO: every active required gate is proved in one compatible scope and the actual endpoint has strict positive slack.
- (ii) ARCHITECTURE-INFEASIBLE: a theorem-backed exact cover of the declared route universe stops every route.
- (iii) REROUTE: the selected route is stopped and a distinct typed alternative with a fresh registry has been selected.
- (iv) STOP-ROUTE: the selected route has a complete actual-carrier stop certificate and no typed alternative is selected.
- (v) NOT-TESTABLE: an active required artifact or type is absent.
- (vi) ARITHMETIC-FRONTIER: every active structural node and the strict physical endpoint are proved, while every unresolved active node is status-OPEN, scope-compatible and explicitly a positive L2 target.
- (vii) OPEN: every remaining valid case.

Theorem 4.2 (Total deterministic classifier). *Manifest validation followed by the displayed precedence returns exactly one of*

INVALID-SNAPSHOT, GO, ARCHITECTURE-INFEASIBLE, REROUTE,
STOP-ROUTE, NOT-TESTABLE, ARITHMETIC-FRONTIER, OPEN.

Proof. Validation either fails, giving the first outcome, or supplies a well-typed route universe, selected route, gate table and endpoint. On a valid snapshot, [definition 4.1](#) is an ordered list ending in a catch-all. The first applicable predicate is unique by precedence. The order deliberately separates global stop cover, selected-route stop, missing data and the arithmetic frontier. $\square$

Remark 4.3 (Conditional synthesis is not a verdict). The existence of [\(1\)](#) is metadata: it tells us what would follow from a complete certificate. It is neither a weakened GO nor evidence that the remaining inputs are technical.

5 The strict endpoint and stop directions

For the MVP1 contract the physical target is

$$\Lambda_{\text{phys}} < \frac{1}{400}$$

on the amplitude scale. The TPC-141 occurrence registry keeps energy-to-amplitude conversion, joint replacement and packet census inside this ledger.

Proposition 5.1 (Upper failure is not a lower obstruction). *For one fully specified route cell:*

- (a) *a complete theorem-backed upper certificate $U < 1/400$ passes the physical endpoint;*
- (b) *a complete theorem-backed lower or exact dual certificate $L \geq 1/400$ stops the route cell; and*
- (c) *a conservative upper certificate $U \geq 1/400$, without such a lower certificate, merely fails to prove the endpoint and does not establish STOP-ROUTE.*

Unknown tokens give NOT-TESTABLE, not zero. H5 determinant reserve cannot be subtracted from U.

This distinction is necessary because a poor upper estimate may be improvable, while a lower certificate excludes every allowed realization of the selected cell.

6 H9 must be arithmetically independent

The physical occurrence registry is a structural certificate. In the MVP4 proof DAG,

$$\text{H9.physical_registry} \leftarrow \{\text{H1.actual_carrier}, \text{H6.cover}, \text{H7.fixed-h}_0, \text{H8.reconnection}\},$$

with no direct or indirect dependency path from H2, H3, H4 or H5.

Proposition 6.1 (Noncircular arithmetic-frontier test). *If H9 is proved using a target H2 or H3 saving, a complete H4 arithmetic tail estimate, or H5 determinant reserve as a physical token, the snapshot cannot be labeled ARITHMETIC-FRONTIER.*

Proof. The arithmetic-frontier predicate is intended to assert that all nonarithmetic interfaces are already closed before the remaining arithmetic targets are attacked. Using one of those targets to prove the physical registry assumes a downstream premise and makes that predicate circular. TPC-141 therefore rejects such an H9 parent edge. $\square$

7 Projection of TPC-133–141

The selected route is `signed_native_frontier_augmented`. TPC-135 stops the `eligible_only_tpc17_18_compiler`, and TPC-138 stops the `direct_quantitative_shift1_reparameterization`. Neither witness stops the selected frontier-augmented route, so the declared route universe has no complete stop cover.

8 First missing and current verdict

The TPC-141 topological scan returns

```
node = H1.frontier_totalization,
status = NOT-TESTABLE,
artifact = total downstream frontier maps or a complete frontier theorem.
```

Table 1: MVP4 gate projection in the frozen snapshot.

Gate	Status	Evidence target	First unresolved content
H1	NOT-TESTABLE	L1	Frontier totalization after the proved cut archive.
H2	OPEN	positive L2	Signed replacement for the unresolved resonance sector.
H3	OPEN	positive L2	Actual CRT containment, squarefree/periodic reassembly, local exceptional-window control, selector/all-prefix and positive fixed- X -power raw-saving theorem.
H4	OPEN	positive L2	Complete original-scale physical high/ultra/boundary return.
H5	NOT-TESTABLE	positive L2	Actual determinant/zero pair on totalized fibers.
H6	NOT-TESTABLE	L1	Physical cover including every frontier leaf.
H7	NOT-TESTABLE	L1	Total fixed- h_0 map on every terminal leaf.
H8	NOT-TESTABLE	L1	Final hard-packet intertwining beyond the cut.
H9	NOT-TESTABLE	L1	Complete occurrence registry and strict physical primal.

Theorem 8.1 (MVP4 route decision). *For the source-locked TPC-133–141 snapshot dated 27 July 2026,*

$$\boxed{\text{Verdict}_{133:141} = \text{NOT-TESTABLE.}}$$

The result is neither GO, nor ARITHMETIC-FRONTIER, nor ARCHITECTURE-INFEASIBLE.

Proof. The source manifest passes validation, so INVALID-SNAPSHOT does not apply. The selected route is not stopped and the route universe is not completely covered by stop witnesses, so the three stop outcomes do not apply. The required frontier-totalization node is absent; hence NOT-TESTABLE occurs before ARITHMETIC-FRONTIER and OPEN. Independently, H6–H9 and the strict endpoint are incomplete, and no positive L2 theorem is present, ruling out both GO and ARITHMETIC-FRONTIER. $\square$

Corollary 8.2 (Precise progress statement). *The first-failure pointer has moved from a generic missing native archive to the first unsupported carrier after an exact cut. This is structural L1 progress. It is not a positive L2 arithmetic event.*

9 Publishable outcomes and next branch

The batch has publishable content even though the route verdict is NOT-TESTABLE:

- executable native enumeration and exact cut conservation;
- a scoped formal coefficientwise obstruction to the eligible-only compiler on the support-envelope archive;
- a fixed-data logarithmic arithmetic closure;
- a restricted small-polylog affine almost-scale power-of-log arithmetic shadow;
- a scoped non-transfer from quantitative shift one;
- exceptional-scale and selector firewalls; and
- a source-locked integration theorem with a directional endpoint calculus.

A bare missing file is not a negative theorem. The publishable content is the exact theorem that reaches and classifies the missing frontier, together with the two scoped route obstructions.

The next minimal branch is now forced:

1. enumerate every frontier terminal from TPC-136;
2. attach Q_D, Q_Z, G and fixed- h_0 maps, or prove the complete frontier sum $o(X)$;

3. rebuild H6–H9 and the occurrence registry without inheriting old slack;
4. certify $\Lambda_{\text{phys}} < 1/400$ independently of H3; and
5. only then test whether all remaining unresolved nodes are actual positive-L2 targets.

If the frontier totalizes and the physical endpoint passes, but H2–H5 remain the only open compatible arithmetic targets, a new frozen snapshot may legitimately return ARITHMETIC-FRONTIER. If an actual kernel or endpoint dual stops the frontier-augmented route, the correct output is STOP-ROUTE or REROUTE, not a claim about twin primes.

10 Deterministic regression and claim boundary

The standard-library script `experiments/tpc142_mvp4_route_audit.py` reads the frozen TPC-141 manifest and checks:

1. manifest hash, source-lock summary, DAG and first-missing consistency;
2. direct and indirect independence of H9 from H2–H5;
3. reachability of all seven valid-snapshot verdicts;
4. rejection of a malformed snapshot before verdict selection;
5. rejection of conditional, frozen and refuted pseudo-frontiers;
6. refusal to infer architecture infeasibility from an incomplete route cover;
7. refusal to turn an upper-bound failure into a stop; and
8. absence of a proved positive-L2 record.

Default mode writes a frozen snapshot and audit JSON. `-check` performs no writes and compares both artifacts byte for byte.

Established. Snapshot validation, ordered verdict selection, the first-missing projection, H9 independence and endpoint direction rules are L0/L1 audit results.

Not established. No complete frontier theorem, H1–H9 conjunction, strict physical endpoint, positive fixed- h_0 L2 saving, $B_{h_0,\delta} = o(X)$, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem is proved.

11 Conclusion

MVP4 records real progress without changing the arithmetic claim boundary. The native archive has become executable through a complete declared cut, not a totalized full carrier, and a false eligible-only compiler has been stopped. Known Chowla information has been placed in its exact frozen, shift-one, or restricted small-polylog almost-scale scope. The remaining structural failure is now one named frontier-totalization node, not a vague missing dictionary.

Because that node and the physical registry are incomplete, the route is not yet at the arithmetic frontier. The correct verdict is NOT-TESTABLE. This verdict is useful precisely because its source, scope, first missing artifact, scoped negative results and permitted next route are all frozen and machine-auditable.

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